

AVIATION CUSTOMER EXPERIENCE & SERVICE ANALYTICS

1. Project Overview and Objective

This project focuses on analyzing airline customer reviews to understand customer satisfaction, service performance, recommendation patterns, and trends in airline customer experience. Using the **Skytrax Airline Review Data**, the project applies Python-based data cleaning, preprocessing, exploratory data analysis, and visualization to transform customer review data into meaningful analytical insights.

The analysis focuses on airline reviews from **2021–2025** to provide insights into recent customer experiences. The cleaned and analyzed data will subsequently be used for Power BI dashboard development to present key customer satisfaction and service performance metrics. The primary objective is to identify patterns in customer ratings, evaluate different service dimensions, compare customer experiences across airlines and traveller categories, and identify service areas that may influence overall satisfaction and recommendation behavior.

2. Data Sources

- **Source Description and Timeline:** Airline Travel Reviews Data (Skytrax), covering airline customer reviews from **2015–2025**.
 - **Original Dataset Source:** Mendeley Data – Airline Travel Reviews Data (Skytrax)
 - **Dataset Used for Analysis:** Reviews from **2021–2025**
 - **Analysis Dataset Size:** **21,984 records × 19 features**
 - **Domain:** Aviation / Airline Customer Experience
 - **Dataset Format:** CSV
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3. Problem Statement

- To analyze airline customer reviews across different airlines and traveller categories to identify patterns in customer satisfaction and service performance.
- To evaluate customer ratings across key service dimensions such as seat comfort, cabin staff service, food and beverages, ground service, inflight entertainment, Wi-Fi/connectivity, and value for money.
- To analyze recommendation patterns and understand how overall ratings and service experiences vary across different customer segments.
- To identify trends in airline customer experience from **2021–2025** and highlight service areas that may require improvement.

- To compare customer experiences across airlines, traveller types, and seat types to generate meaningful business insights.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
review_date	Date	Date on which the customer review was posted
over_all_rating	Numeric (Integer)	Overall customer rating given to the airline on a scale of 1–10
review_header	String (Text)	Title or heading of the customer review
trip_verified	Categorical	Indicates whether the review is verified
review_text	String (Text)	Detailed textual feedback provided by the customer
aircraft	String (Text)	Aircraft information associated with the review; excluded during preprocessing due to high missing and inconsistent values
type_of_traveller	Categorical	Type of traveller, such as Business, Couple Leisure, Family Leisure, or Solo Leisure
seat_type	Categorical	Type or class of seat used by the customer
route	String (Text)	Flight route mentioned in the review
date_flown	Date / Text	Date or period when the flight was taken
seat_comfort	Numeric	Customer rating for seat comfort on a 1–5 scale
cabin_staff_service	Numeric	Customer rating for cabin staff service on a 1–5 scale
food__beverages	Numeric	Customer rating for food and beverage services on a 1–5 scale
inflight_entertainment	Numeric	Customer rating for in-flight entertainment on a 1–5 scale
ground_service	Numeric	Customer rating for ground services on a 1–5 scale
wifi__connectivity	Numeric	Customer rating for Wi-Fi and connectivity services on a 1–5 scale
value_for_money	Numeric	Customer rating for value for money on a 1–5 scale
recommended	Categorical	Indicates whether the customer recommends the airline
airline_name	Categorical	Name of the airline reviewed

5. Tools & Technologies

- **Python:** Data loading, cleaning, transformation, exploratory data analysis, and visualization.
- **Pandas:** Data manipulation, missing-value analysis, filtering, and transformation.
- **NumPy:** Numerical operations and statistical calculations.
- **Matplotlib:** Data visualization and customized charts.
- **Seaborn:** Statistical visualization and exploratory analysis.
- **Power BI:** Subsequent data modelling, DAX calculations, visualization, and interactive dashboard development.

- **Google Collab:** Python development and notebook-based analysis.
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6. Data Pre-Processing - EDA

Tasks Performed:

- **Import Libraries** – Import Pandas, NumPy, Matplotlib, Seaborn, and Warnings.
 - **View First 5 Rows** – Use `df.head()` to understand the data.
 - **Check Dataset Size** – Use `df.shape` to check rows and columns.
 - **Check Data Types** – Use `df.info()` to understand column types and structure.
 - **Statistical Summary** – Use `df.describe()` to view numerical statistics.
 - **Check Missing Values** – Use `df.isnull().sum()` to identify missing values.
 - **Check Duplicate Records** – Use `df.duplicated().sum()` to identify duplicates.
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